

Carolina Brañas

Data Scientist | Machine Learning & Geospatial Data

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Data scientist with an M.Sc. in Social Data Science and two years of research experience at the University of Copenhagen. Experience with machine learning and geospatial data (raster and vector processing with Rasterio, GeoPandas and QGIS). Research experience helped me become more creative by taking open-ended problems, framing them and finding solutions that work.

RESEARCH INTERESTS

Deep learning, environmental data and climate change, geospatial data, self-supervised learning (previous experience with clustering techniques), and the development of new global-scale datasets.

EDUCATION

M.Sc. Social Data Science

2022 – 2024

University of Copenhagen, Denmark

Thesis: **Mobility and income segregation in Madrid, Spain.** Used GeoPandas to join Madrid's district geometries to income data, derive inter-district distances for origin–destination matrices from a month of MITMA trip records, build the spatial weights for Global and Local Moran's I, and map segregation patterns; plus network analysis of the mobility graph.

Relevant Courses: Advanced Machine Learning, Geospatial Data Science, Advanced Network Science (ITU); NLP (DIKU).

B.Sc. Marketing, Computer Science Minor

2018 – 2022

Cardinal Stritch University, United States

Honours: Magna Cum Laude, Dean's List every semester, best graduating GPA of Marketing B.Sc. (2022), full four-year academic and athletic scholarship.

Relevant Courses: Object-Oriented Programming, Computer Architecture, Structured Programming, Introduction to Artificial Intelligence.

RESEARCH & PROFESSIONAL EXPERIENCE

Data Science Research Assistant *University of Copenhagen, Denmark*

Feb 2026 – Present

↔ Prof. Asger Mose Wingender

- Building a new dataset for Cataloguing Crop Traits and Breeders Across the OECD 1980–2025 (**Carlsberg Foundation**): the first unified record of every cereal variety released in the OECD since 1980, with traits, breeding history and the companies behind them.
- Own the data model and ingestion workflow consolidating heterogeneous national registries into one analysis-ready dataset; manage a student assistant on the project, coordinating and reviewing their work.

Data Science Research Assistant *University of Copenhagen, Denmark*

May 2024 – Dec 2025

↔ Prof. Jeanet Bentzen

- Contributed to the **Shocking Religion** project on religion's economic impact, in a multidisciplinary team.
- Computed distances from locations on Earth to potential earthquake danger areas for each of the four hazard levels (1–4): rasters in Rasterio, nearest-distance operations in GeoPandas, mapping in QGIS.
- Built topic models over large text corpora and RAG-based LLM pipelines for document insights.
- Containerized scalable data workflows (Docker) and ran large-scale ingestion and preprocessing on cloud infrastructure (UCloud, GPU compute).

Data Scientist *Above Sports, Denmark*

Oct 2023 – May 2024

- Developed deep learning computer vision models for brand logo detection in sports footage.
- Automated and dockerized data workflows for reproducible training and inference runs.

Marketing Strategist *Crescendo Collective, United States*

Sep 2021 – May 2022

- Automated internal reporting with Python and analysed campaign and audience data.

SKILLS

Programming & Data: Python (pytorch, tensorflow, scikit-learn, pandas, numpy, matplotlib), SQL, Bash, L^AT_EX

Machine Learning: deep learning, computer vision, Transformers, topic modeling (UMAP, HDBSCAN, BERTopic), LLMs & RAG pipelines, Hugging Face, W&B

Geospatial & Remote Sensing: Rasterio, GeoPandas, QGIS, raster & vector processing, spatial joins, network analysis

Cloud & DevOps: Docker, Git, Linux, AWS, Azure, UCloud (GPU compute), CI/CD

Languages: Spanish (native), Galician (native), English (professional proficiency), Danish (beginner)

REFEREES

Currently waiting for references due to the summer break; these will be provided after the application deadline.